Pre−Pandemic Data Only

- antiGen
- EVEscape
- ESM 2 (35M)
- ESM 2 (650M)
- BLOSUM62
- Codon table

antiGen Finetunes

- antiGen
- antiGen as ESM 2 (35M) LM head FT
- antiGen as ESM 2 (35M) full FT
- antiGen as ESM 2 (35M) partial FT
- antiGen as ESM 2 (650M) LM head FT
- antiGen as ESM 2 (650M) full FT
- antiGen as ESM 2 (650M) partial FT
- antiGen as EVE full FT
- antiGen as Tranception full FT

Pandemic Data in Training Window

- antiGen
- PSSM
- PSSM Conservation
- MSA Transfomer
- MSA Trans. (BA.2 eval)
- EVE
- Tranception
- Substitution matrix
- Where × which

Deep Mutational Scans

- antiGen
- BA.1 DMS (bind)
- BA.1 DMS (expression)
- BA.2 DMS (bind)
- BA.2 DMS (expression)
- BA.2 DMS (LY−CoV1404 escape)

Ensembles

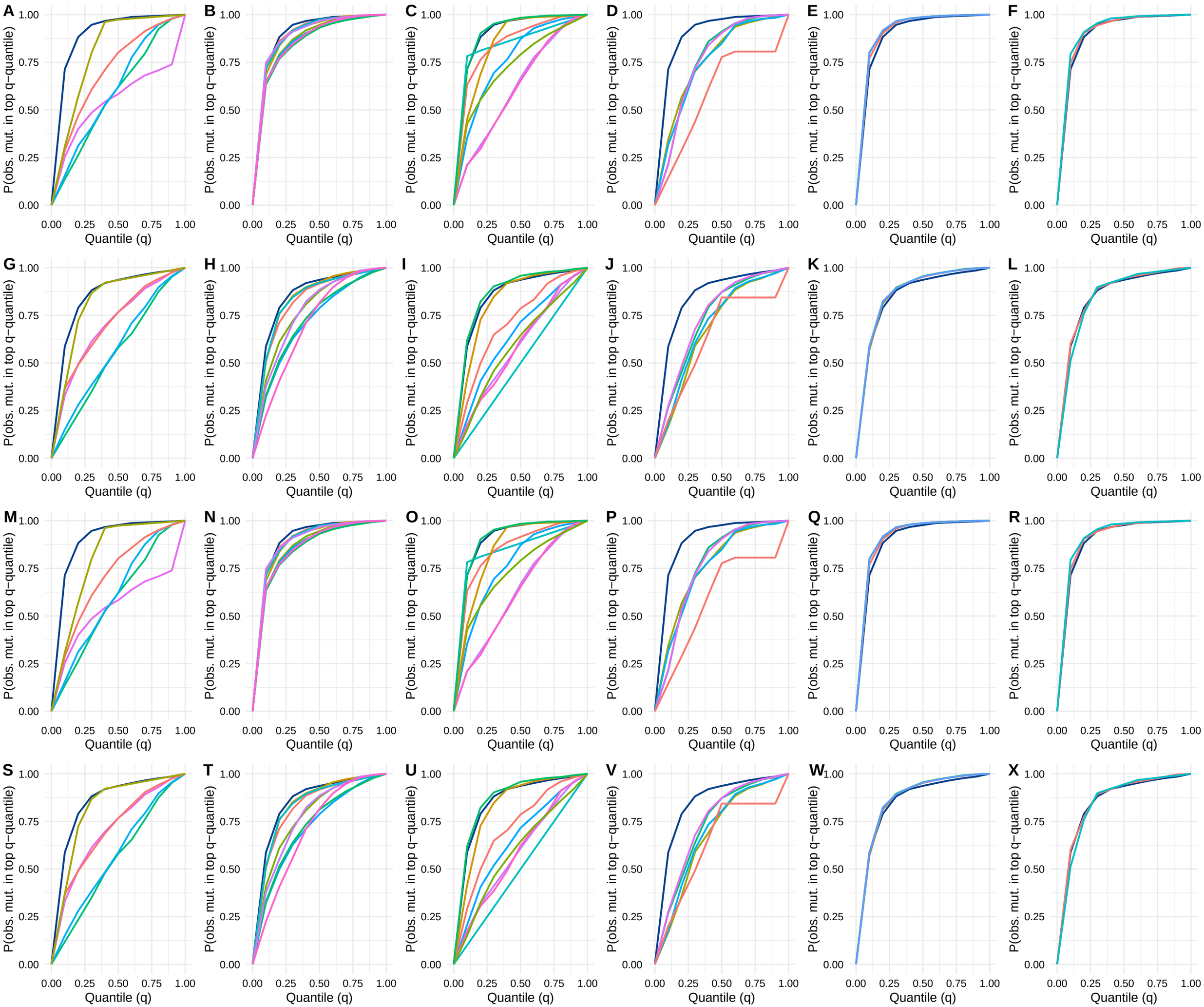
- antiGen
- Mean of ranks
- Mean of log−probs
- Mean of z−scores

Val−selected, averaged over 10 holdouts

- antiGen
- antiGen, val−selected (10x)
- Where x which + PSSM, val−selected (10x)

all mutations

de novo

gap/stop mask

de novo + gap/stop mask